

Finance Q&A with Named Entity Recognition (NER)

Ask questions and see named entities

(companies, names, addresses)

highlighted in the response.

Screenshots:

1 explain ****GreenTech Inc. Sustainability Financial Statement****

The screenshot shows a web application titled "Finance Q&A with Named Entity Recognition". It features a search bar with the prompt "Ask a question about finance or business" and a text input field containing the question "explain 'GreenTech Inc. Sustainability Financial Statement'". Below the input field are "Clear" and "Submit" buttons. To the right, the application displays the response text, which is a detailed sustainability financial statement for GreenTech Inc. The response includes sections for Environmental Metrics (Carbon Emissions, Energy Consumption, Water Usage), Social Metrics (Employee Diversity, Community Engagement), and Governance Metrics (Board Diversity, Ethics and Compliance). Below the response, a box labeled "Entities Found:" lists the named entities: GreenTechInc. (COMPANY), GreenTechInc. (COMPANY), GreenTechInc. (COMPANY), and GreenTechInc. (COMPANY). At the bottom, there is a "Flag" button and a section for "Examples" with several sample questions.

2 what info you know about Dear Lorraine Fischer-Maurice

The screenshot shows the same web application with the question "what info you know about Dear Lorraine Fischer-Maurice" entered in the text input field. The response text provides information about Lorraine Fischer-Maurice, including her role as a policyholder, her insurance policy details (policy number C225-2503-487, premium due of \$500), and her coverage details (Personal Liability Coverage, Property Damage Coverage, Medical Payments Coverage). The "Entities Found:" box lists: LorraineFischer-Maurice (NAME) and Lorraine (NAME). The "Examples" section at the bottom remains the same.

3 what info you know about Title: Network Outage - Angelika Stroh

The screenshot shows the same web application with the question "what info you know about Title: Network Outage - Angelika Stroh" entered in the text input field. The response text describes a network outage affecting Angelika Stroh's workstation at 1:19 PM on January 15, 2023, and the steps taken to resolve the issue. The "Entities Found:" box lists: 1/15Miguel-Vollbrecht-Gasse (STREET ADDRESS) and AngelikaStroh (NAME). The "Examples" section at the bottom remains the same.

4 Tell me about Microsoft's CEO Satya Nadella.

Finance Q&A with Named Entity Recognition

Risk questions and see named entities (companies, names, addresses) highlighted in the response.

Ask a question about finance or business

Tell me about Microsoft's CEO Satya Nadella.

Clear Submit

I don't have specific information about Satya Nadella in the provided context.

Entities Found:

NAME SatyaNadella

Flag

Examples

explain "GreenTech Inc. Sustainability Financial Statement" | what info you know about Dear Lorraine Fischer-Maurice | what info you know about Title: Network Outage - Ringelka Strah

Tell me about Microsoft's CEO Satya Nadella. | What companies did John Smith work for? | When Cameron-Mcknight Supply Chain Management Agreement entered into place?

5 What companies did John Smith work for?

Finance Q&A with Named Entity Recognition

Risk questions and see named entities (companies, names, addresses) highlighted in the response.

Ask a question about finance or business

When Cameron-Mcknight Supply Chain Management Agreement entered into place?

Clear Submit

John Alexander Doe, as per the provided information, worked for two companies: 1. ABC Corporation Ltd as a Software Engineer for 3 years. 2. XYZ Ltd as a Software Developer from 2015 to 2018.

Entities Found:

COMPANY ABCCorporationLtd

NAME JohnAlexanderDoe

COMPANY XYZLtd

Flag

Examples

explain "GreenTech Inc. Sustainability Financial Statement" | what info you know about Dear Lorraine Fischer-Maurice | what info you know about Title: Network Outage - Ringelka Strah

Tell me about Microsoft's CEO Satya Nadella. | What companies did John Smith work for? | When Cameron-Mcknight Supply Chain Management Agreement entered into place?

6 When Cameron-Mcknight Supply Chain Management Agreement entered into place?

Finance Q&A with Named Entity Recognition

Risk questions and see named entities (companies, names, addresses) highlighted in the response.

Ask a question about finance or business

When Cameron-Mcknight Supply Chain Management Agreement entered into place?

Clear Submit

The Supply Chain Management Agreement between Cameron-Mcknight and Vendor was entered into on the 1st day of March, 2021.

Entities Found:

COMPANY CameronMcknight

COMPANY Vendor

Flag

Examples

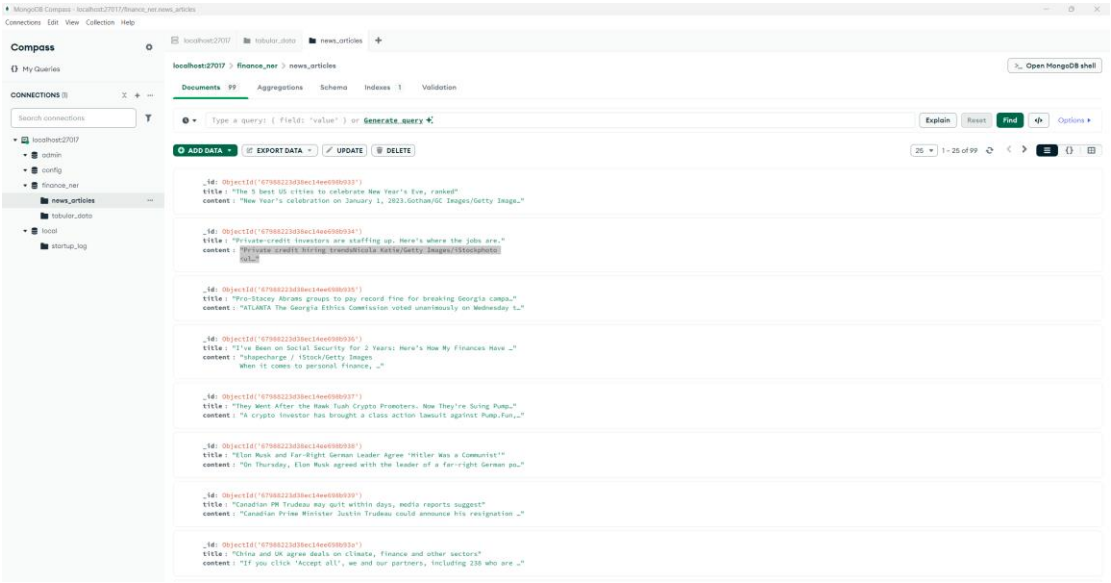
explain "GreenTech Inc. Sustainability Financial Statement" | what info you know about Dear Lorraine Fischer-Maurice | what info you know about Title: Network Outage - Ringelka Strah

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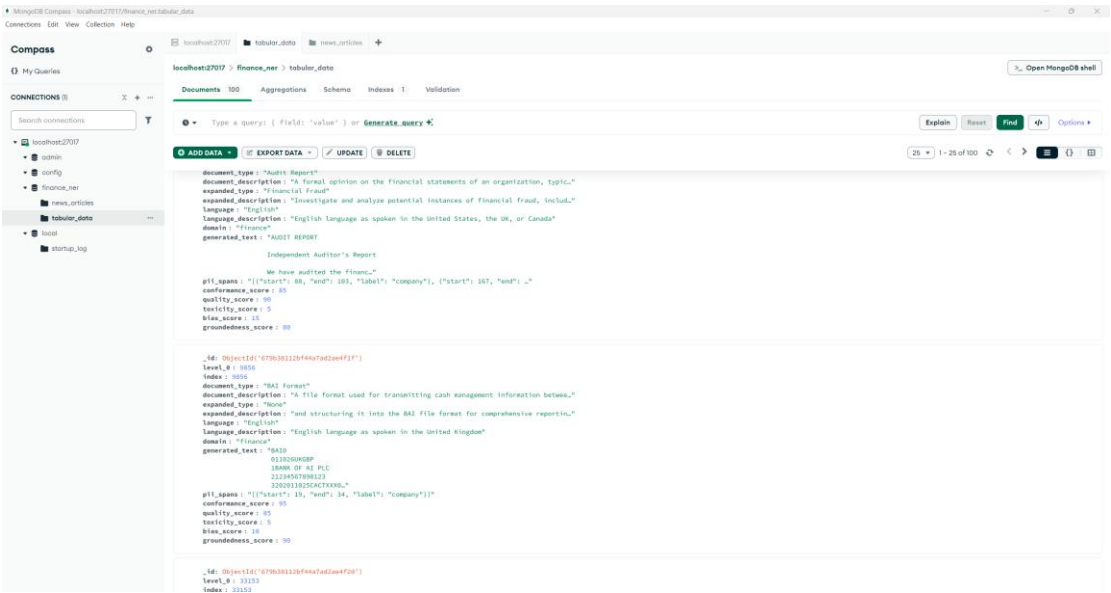
*Examples questions with answers taken from MongoDB.

MongoDB Structure:

1 news_articles



2 tabular_data



NER Explained:

The "B-" and "I-" prefixes in the label system are part of the BIO (Beginning, Inside, Outside) tagging scheme, which is a standard format used in Named Entity Recognition (NER) tasks. Here's what each prefix means:

B- (Beginning): Marks the first token of a named entity

Example: In "John Smith", "John" would be tagged as "B-PERSON"

Indicates the start of a new entity

I- (Inside): Marks subsequent tokens that are part of the same entity

Example: In "John Smith", "Smith" would be tagged as "I-PERSON"

Shows continuation of the entity that started with B-

O (Outside): Marks tokens that are not part of any named entity

Example: In "John lives in London", "lives" and "in" would be tagged as "O"

text

Copy

Text: "John Smith transferred \$5000 to Bank of America"

Labels:

- John

-> B-PERSON

(start of person name)

- Smith

-> I-PERSON

(continuation of person name)

- transferred

-> O

(not part of any entity)

- \$5000

-> B-MONEY

(money entity)

- to

-> O

(not part of any entity)

- Bank

-> B-ORG

(start of organization name)

- of

-> I-ORG

(continuation of organization name)

- America

-> I-ORG

(continuation of organization name)

The BIO scheme is particularly useful because it:

1. Helps distinguish between adjacent entities of the same type

2. Makes it clear where entities begin and end

3. Allows for precise entity boundary detection

4. Is a standard format that many NLP tools and models expect

Without the B- and I- prefixes, it would be harder to determine whether two adjacent tokens with the same entity type are part of the same entity or represent different entities.